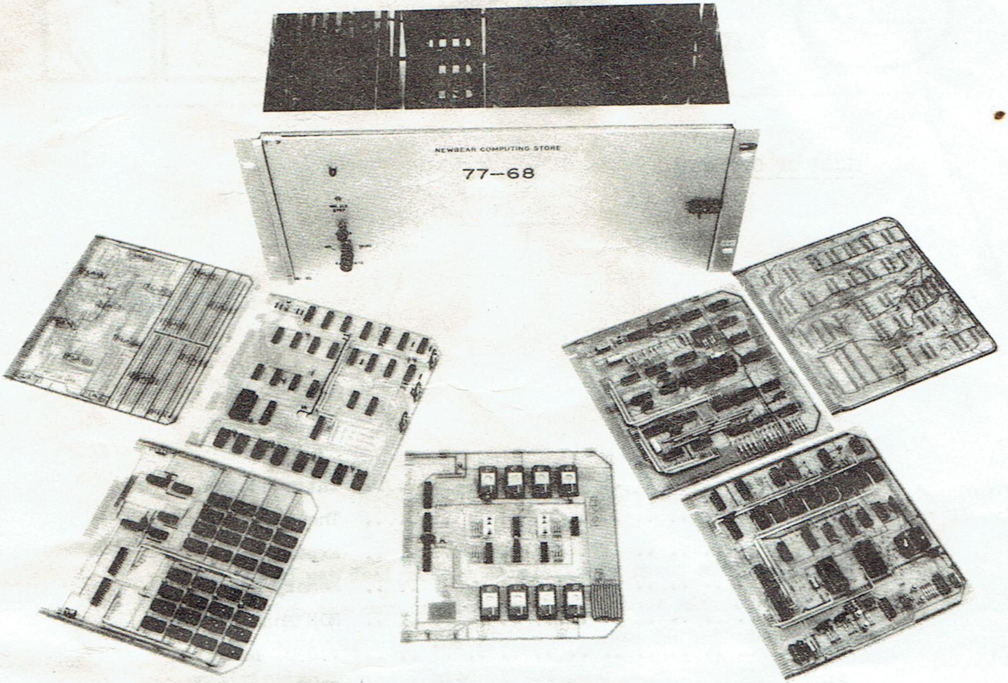

THE 77— 68 SYSTEM

An. Introduction...

BY

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NEWBEAR COMPUTING STORE.

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77-68.

1.) Introduction.

The 77-68 was designed as a relatively cheap and simple way with which the constructor can learn about microcomputing by direct experience. The system can then be expanded without restriction to the limits of the 6800 microprocessor's potential, and at the same time allowing ample flexibility for the system to be structured to suit the constructors individual requirements.

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2.) The Basic Machine.

The 77-68 is designed around the Standard 8" square printed circuit board. The basic machine comprises of a single board containing the 6800 microprocessor chip, 256 bytes of memory, data input and output registers and miscellaneous control circuitry (Bear Bag No.1).
A simple control panel - address and data entry switches and LED indicators. (Bear Bag No 2).
A source of 5 volts D.C. @ about 1amp (Bear Bag No.3).
Even in this simple form the 77-68 can be used to execute real programmes. Mathematical routines can be run, simple computer games played or the machine can be a controller/ sequencer for household devices.
However a word of warning, although the 77-68 is relatively straightforward it should not be attempted until the would-be constructor is familiar with the construction of TTL based logic circuits, and has some knowledge of microcomputing hardware and software.
Also he should be fairly competent with a soldering iron.
The following books are highly recommended as suitable background reading:-

1. An Introduction to Microcomputers Vol.I & II. (by Adam Osbourne).

2. M6800 Microcomputer Systems Design Data. (or at least the M6800 data sheets).

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3.) Expansion.

This board has been produced as a possible first stage expansion to the basic machine.

The Mon.1 board provides the facilities for:-

a) Linking up to an external programme storage device eg. Audio tape recorder or Paper tape reader etc.
b) Connecting up a V.D.U. or teletype for input and output.
c) Implementing some form of Monitor, or simple operating system for driving the I/O device.

The Mon.1 board is a single 8" square board, with fully buffered data and address lines, and is in all ways compatible with 77-68 system bus.
Power requirements are:-

+5v DC.Stabilized @ 0.55A.
+12v DC " @ 100mA.
-12V DC " @ 50mA may be required depending on type of I/O used.

Serial I/O provided on board is max of 2 Transmit and 2 receive asynchronous serial ports using standard data transmission rates from 50 to 9600 Baud. (24) or 20mA (teletype) I/O provided for. Also 1024 Bytes of RAM which may be write protected, are provided plus a 32 Byte Bootstrap ROM.

3). Expansion/cont.

Available to run in conjunction with the Mon 1 is 'Minimon' a 1k Byte soft monitor on cassette tape which can be loaded into the 1k RAM on Mon 1. via the Bootstrap Loader. This monitor contains the usual features, such as Exam/Modify memory:- Set Break Point:-Go to/Run user programme, and also the very useful feature of calculating relative offset (for BNE, BEQ etc., Type instructions).

4). Now for some memory.

Two memory boards are available:-

A 4kRam Board (Bear Bag 5). This board provides a 4096 word random access memory for use with the 77-68 system, Board size and pin connections being as per other 77-68 system cards. The board uses 2102 or 2102-1 memory chips and can be initially equipped with only 1k of memory, and expanded in 1k increments as funds allow.
Power requirements are +5vDC stabilized @ between 1.1amp and 1.8amp depending on memory chips used.

5). Also available is a 32k dynamic memory board.

This uses 4116 dynamic memory chips, the memory being organised in 2 16k blocks. All refresh control logic being on board. An on-board 8 way DIL switch allows each of the 16k blocks of memory to be positioned anywhere within the 16 Bit address space, also one 16k block may be write protected if required. Power requirements being $\pm$ 12vDC and +5vDC.

6). For permanent storage of user programmes on cassette there is available:-

a) Bear Bag No.10.'Kansas City Cassette' interface board which uses the RS232C (V24) serial I/O between the Computer and Terminal.
b) Bear Bag No.18 'Cottis Blanford Cassette' interface which will interface directly with a ACIA or UART.

Both of these cassette interface boards work perfectly from an ordinary domestic cassette recorder.

So far we have a system which comprises of a CPU board, a monitor board, some memory and a cassette interface which requires either a RS232(V24) terminal or a Teletype (20A current loop) for communication. However there is a second choice available.......

7). ROM MONITOR BOARD (Bear Bag No.13).

The Rom Monitor Board makes available to the user a monitor programme in ROM. either MIKBUG or the updated version SWATBUG, alternativly a 2708 can be used with the users own custom monitor in it; or the 2708 facility can simply be used to store some frequently used user programme.
An additional PIA has been included giving 16 I/O ports for monitoring and/or parallel tape loading etc., also to aid programme development a single step facility is included.
The main features of the Mon 2 Board are:-

* Maintains all the facilities of the 77-68 CPU.
* Buffered address and data lines.
* MIKBUG/SWATBUG monitor ROM.
* Programmable timer for V24 serial interfacing, signalling speeds can be preset to 110, 300 or 1200 baud.

Main features of the Mon 2 Board/cont...

* PIA used by MIKBUG/SWATBUG for serial interfacing.
* ACIA at address 8000_{16} for user programming; can be moved to 8004_{16} for high speed serial interfacing with SWATBUG.
* Access to the 256 bytes of scratch-pad memory located on the 77-68 CPU PCB at base address $A000_{16}$
* Single step facility which can display the contents of the CPU's registers after each programme step.
* 1k byte of EPROM located at base address $E400_{16}$

8). V.D.U. Board.(Bear Bag No.9)

This Visual Display Unit board gives the 77-68 user fast and silent communication with his computer. It drives a standard television set or monitor giving an upper and lower case alphanumeric display of 24 lines or 40 characters, and will accept input from a separate alphanumeric keyboard.
By designing the board as part of the 77-68 computer, it has been possible to treat the display refresh memory as a part of the main computer memory. Thus each character position on the screen corresponds to a particular memory location, and can be individually written into and read from by the CPU. The screen contents can therefore be changed very rapidly, and also any individual character can be changed without affecting the rest of the display. Individual characters may be displayed as white on black background, or as black on white background. This feature is useful for drawing attention to certain parts of the display, for differentiating between human input and computer output; or for giving a limited 'graphics' capability by treating the screen as a 24 x 40 matrix of black or white rectangles.

V.D.U. Board/Cont...

The printed circuit board available for the VDU also contains a small 'prototyping' area for those who may wish to try different character generators or add other circuitry.

9). To tie Mon 2 and the V.D.U. board together we have the TBUG software package.

TBUG has been written for the 77-68 system with MON 2 and the 77-68 mapped VDU. So MON 2 is compatible with the

Board Characteristics

Construction:-	8.0" x 8.0" single sided PCB with gold plated edge connector contacts.
Bus:-	Buffered, 77-68 CPU compatible.
Power:-	Requires +5vDC stabilized @ 0.25A typical.
I/O Lines:-	32 parallel data lines, programmable as inputs or outputs. 8 parallel I/O control lines. 3 timer clock inputs. 3 timer gate inputs. 3 timer outputs.
Addressing:-	On board switches select one of 16 address 'blocks' within range F608-F7FF. allowing mutiple PIO-A boards to be fitted in a system. Compatible with 77-68 MON I. address allocation.

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Main features of TBUG/Cont...

* Commercial software (e.g. basic games etc.,) runs without modification.
* So now the advantages of a memory mapped VDU need not be forfeited, to use a MIKBUG compatible monitor.
* Single step trace with unmodified hardware on Mon-2 board.
* Single step facility which can display the contents of the CPU's registers after each programme step.
* 1k byte of EPROM located at base address $E400_{16}$

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10).

Further enhancements to the system are a PIO Board, this is a general purpose Parallel Input/Output board for use with the 77-68 system. It can mount one or two 6821 PIA's and a 6840 Programmable Timer Module, and contains all the necessary system bus buffers and address decoding.
A large 'prototyping' area is provided on the board for the users particular interfacing circuits, and two 'scotchflex' 40 way connectors which the user does not wish to route via the spare contacts of the board 78-way edge connector.

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11). <u>and a ROM Board</u>.

This board allows the 77-68 user to add EPROM IC's (2708 1k x 8 or 2716 2k x 8) to his system. Two blocks of four EPROM sockets are provided, each 'block' may be set up to take either four 2708's or four 2716's.

The address of each 'block' may be independantly selected to be any 4k (or 8k for 2716's) segment within the 77-68's 64k byte addressing range.

Multiple boards may be fitted to a system as required.

Board Characteristics

Capacity:-	Up to 8 2708's (8k bytes) or 8 2716's (16k bytes) or 4 2708's + 4 2716's (12k bytes)
Bus:-	77-68 compatible
Speed:-	Depends on EPROM fitted; compatable with 1.6uS 77-68 CPU even with low speed 2708's.
Power:-	Requires + 5v stabilized @ typically 0.1A plus 6mA/2708, 10mA/2716 fitted. Also +12v stabilized @ 50mA/2708 -12v stabilized @ 30mA/2708 (+ & -12v not required for 2716's)
Construction:-	8.0" x 8.0" single-sided PCB with gold-plated 0.1" edge connector pads

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12).

For those wishing to use a Standalone (RS232) V.D.U.
We have PETITEVID (Bear Bag No.9.)

The NewBear "Petitevid" is a single-board 'electronic typewriter' video controller based on the Thompson-CSF SFF 96364 IC. It will cause a suitable TV monitor (composite video input, 625 lines, 50 Hz frame) to display 16 rows of 64 upper case ASCII characters, according to the contents of the on-board 1K (6 bit) memory.
The memory is loaded sequentially via a blinking (2Hz approximately) 'underscore' cursor, which itself is directable by the input serial start/stop-encoded ASCII character stream.
A UART converts the received serial data to its parallel form and will also convert up to 7 parallel bits of strobed data (from, say, a keyboard) to its serial form. Transmitter and receiver work in the same start/stop code format and at the same baud rate.
Serial data line signals are within V24 (RS 232 C) specifications.

Additional features of the Petitevid include:-

* "Clear Screen" (and home cursor to top left of screen) from keyboard or via local switch.
* Automatic scrolling after sixteenth line.
* Home cursor, without clearing screen, from keyboard.
* User-selectable positive/negative video.
* Three independently adjustable and locally switched UART data rates (up to 120 characters per second-line data rate varies with format chosen);
* By on-board modifications, user may select various serial data formats differing from that supplied.
* On-board regulation of "+5v" supply from, say, the now commonly used "+8v" available from micro-computer mainframe.

Power supplies required are:-

Power supplies required/cont...

-12v (±5%) @ 100mA.

0v (Common ground)

+7v To + 12v (for "+5v") @ 600mA.

13.) To complete PETITEVID, we recommend our 63 Key 'ASR 33' style Keyboard Kit, and if the user wishes to utilise his black and white television we can supply a UHF Modulator to convert the video output from Petitevid to that which can be plugged into the ariel socket of a television.

14.) To economically and conveniently hold the 77-68 system we offer a 19" Rack Unit (Bear Bag No.4.) This used a Vero board back-plane, and the advantage of this [redacted] that as edge connectors [redacted] be added as when required, the need to purchase an expensive motherboard removed.

15.) For more detailed information there is available a 77-68 Manual which covers construction and test details, together, with full circuit diagrams and principle of operation of CPU boards, Mon.1. Boards, and 4K Ram Boards.

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16.)

Detailed information on the other Boards available in the 77-68 range are in the form of Design Notes. eg.,

MON.2.	Design Note.38
ROM A.	Design Note.47
PIO.	Design Note.49
32k RAM.	Design Note.53 etc.,

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T B U G.

For 77-68 Mon 2/77-68 V.D.U.

TBUG is a monitor for use with the 77-68 MON.2 board and memory map VDU board. It is supplied on a 2708 EPROM which is placed in the EPROM socket on the MON 2 board.

MIKBUG has few facilities and, in particular, is poorly suited to the 77-68. Other monitors, although having good features, are unable to take advantage of the memory map VDU. TBUG has been developed specifically to match the machine and its facilities and also to retain as much compatibility with MIKBUG as possible so allowing most existing software to run unaltered.

TBUG has the following features:-
- -12 commands including all 5 MIKBUG commands.
- -MIKBUG compatible subroutine addresses.
- -Most software which runs under MIKBUG will run unmodified.
- -Full VDU scrolling of the 40 x 24 character VDU.
- -Graphics on the VDU are made easy with 'plot' subroutine.
- -Cassette file handling.
- -Input/Output on a serial port (ACIA @8000).
- -Input on two 8 bit parrallel ports (PIA @8010).
- -Full documentation and a listing of the monitor code.

To use TBUG the MON.2 board needs a minor modification to move the EPROM address to @E000. The MIKBUG PROM location is then redundant.

TBUG uses the ACIA for input/output and the PIA input/output (used by MIKBUG) is not used.

The single step feature of MON.2 needs modification as the address of the RTI instruction entering a program is not compatable with MIKBUG.

TBUG uses store locations @A080 through @A091 and thus software using this area may not run correctly.

Output, as well as being displayed on the VDU, is output via the ACIA which allows use of TBUG even if there is no VDU.

The cassette file handling system allows output to a cassette to be preceeded by a file name. Subsequently the tape can be searched for a specific file which is then loaded.

TBUG may be obtained from:- Newbear Computing Store Limited,
40, Bartholomew Street,
Newbury, Berkshire. RG14.5LL.

Price.........................£14.00 Plus 15% V.A.T.

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